

Rachana Reddy
Senior Data Engineer
rachanaareddy@gmail.com
(470)-632-6035
[linkedin.com/in/rachanaareddy](https://www.linkedin.com/in/rachanaareddy)

Professional Overview:

- **Google Cloud Certified Senior Data Engineer** with over **11 years of progressive experience** designing, developing, and deploying scalable data engineering solutions across **banking, financial services, and healthcare domains**.
- Proven track record of delivering high-performance real-time, batch, and streaming data pipelines in cloud-native and hybrid Environments, leveraging modern data platforms like **Snowflake, Databricks, GCP, AWS, and Azure**.
- Hands-on expertise in **Apache Spark (PySpark/Scala), SQL, Kafka, Airflow, and Hadoop ecosystem tools (Hive, HDFS, HBase, Impala)**, with strong foundations in data architecture, schema design. Adept at integrating structured and semi-structured data using **Talend, AWS Glue, DBT**, and managing large-scale migrations from **SQL Server, Oracle, and PostgreSQL to Snowflake, Redshift, and BigQuery**.
- Skilled in real-time data streaming, implementing high-throughput **Kafka, AWS Kinesis, and Apache Airflow pipelines** for fraud detection, risk analytics, and financial transaction monitoring. Developed and optimized interactive dashboards using **Power BI, Tableau, and AWS Quicksight**, enabling real-time executive decision-making for financial insights.
- Specialized in query performance tuning and optimization, leveraging Snowflake clustering, partitioning, and materialized views to reduce query execution time by 40%. Strong understanding of security best practices, implementing data encryption, **RBAC, and IAM policies in Snowflake, AWS, and Azure** to ensure compliance with **PCI DSS, GDPR, and SOX**.
- Specialized in query performance tuning and optimization, leveraging Snowflake clustering, partitioning, and materialized views to reduce query execution time by 40%.
- Hands-on experience in deploying **Dockerized** applications on **AWS Fargate and ECS**, streamlining serverless container management for cloud-based data pipelines. Proficient in **Infrastructure as Code (IaC)** automation using **Terraform, AWS CloudFormation, and AWS CDK**, ensuring scalable and reproducible cloud deployments.
- Strong collaboration skills, working closely with **cross-functional teams, data engineers, and business stakeholders** to optimize **SQL Server and Snowflake** performance while maintaining data lineage, metadata management, and compliance standards. Passionate about leveraging data engineering, cloud analytics, and automation to drive business insights, operational efficiency, and scalable data architectures.
- Proficient in **SQL Server, T-SQL, Power BI, SSRS, and Tableau**, with a strong background in **Azure Data Lake Storage (ADLS), Azure Synapse, and Databricks** for cloud-based data solutions.
- Designed and optimized **ETL pipelines** for structured and unstructured data using **Apache Spark, AWS Glue, and Databricks Delta Lake**, reducing execution times by 35%.
- Led **SQL Server to Snowflake and ADLS migrations**, ensuring **seamless data transformation, validation, and governance** for regulatory compliance.
- Developed **real-time data streaming solutions** using **Kafka, AWS Kinesis, and Apache Airflow**, enhancing **low-latency financial transaction processing**.
- Built **high-throughput event streaming pipelines**, reducing message lag and improving **data scalability for fraud detection and risk modeling**.
- Created **interactive dashboards** using **Power BI, Tableau, and AWS Quicksight**, improving **financial risk analytics and executive decision-making**.
- Optimized **Snowflake workloads** with clustering, partitioning, and materialized views, reducing **query execution time by 40%**.
- Implemented **data encryption and role-based access controls (RBAC)** within **Snowflake and AWS IAM**, ensuring **compliance with financial regulations (PCI DSS, GDPR, SOX)**.
- Automated **infrastructure provisioning** using **AWS CloudFormation, Terraform, and AWS CDK**, improving **scalability and operational efficiency**.
- Collaborated with **data engineers, architects, and business stakeholders** to optimize **SQL Server and Snowflake**.

Scholarly Pursuits:

Bachelor of Technology in Computer Engineering, India.
Geethanjali University

2013

Technical Skills:

Category	Tools and Technologies
Programming & Big Data	Python, Java, Scala, Pandas, Numpy, Pyarrow, Pytest, Scikit-Learn, Boto3, PySpark, Hadoop, Apache Spark, Hive, YARN
Database Management	Amazon RDS (MySQL, PostgreSQL, Oracle), Amazon Redshift, SQLAlchemy, Amazon DynamoDB
API & Web Development	FastAPI, Flask, AWS API Gateway, Django
Analytics & Business Intelligence	Amazon Redshift, AWS Glue, Amazon Athena, QuickSight, Tableau, PowerBI
Cloud Data Management	GCP, Amazon S3, Amazon RDS, Amazon Redshift, AWS DynamoDB, AWS Data Lake Formation
Integration & Automation Services	AWS Glue, Amazon Kinesis, Amazon SQS, Amazon SNS, AWS Lambda, AWS Step Functions, AWS EventBridge, AWS Data Pipeline
Real-Time Streaming & ETL	Talend, DBT, Amazon Kinesis, AWS Glue, AWS Lambda, Amazon EMR, AWS Step Functions, MSK (Managed Streaming for Apache Kafka), Firehose
Containerization & Orchestration	Amazon ECS (Elastic Container Service), AWS Fargate, Docker, Kubernetes, Amazon EKS (Elastic Kubernetes Service)
Infrastructure as Code & Core Services	Terraform, AWS CloudFormation, AWS CDK, AWS OpsWorks, Amazon EC2, Amazon S3, AWS EFS (Elastic File System), AWS VPN
Machine Learning & AI	AWS SageMaker, TensorFlow, Boto3, AWS Deep Learning AMIs, Amazon Rekognition, AWS Polly, Vertex AI, Computer Vision, NLP, PyTorch, Keras
Data Visualization & CI/CD	AWS CodePipeline, AWS CodeBuild, GitHub, Bitbucket, AWS CloudWatch, AWS QuickSight, Tableau, PowerBI, Looker
Networking & Security	IAM, VPC, Subnets, Firewall Rules, AWS VPN, AWS Shield, AWS WAF, AWS CloudFront

Certifications

- Google Cloud Professional Data Engineer – Google Cloud
 - Scrum Master (CSM) – Scrum Alliance
 - Agile Project Management– Google
 - Six Sigma Green Belt — Central Michigan University
- March 2025

July 2023

September 2023

March 2022

Work Portfolio:

Citi Bank, Remote
Senior Data Engineer
Functional Role Details:

March 2024 - Present

- Developed **custom data recovery solutions** leveraging **Snowflake’s Time Travel and Fail-safe**, ensuring data integrity throughout migration.
- Designed and orchestrated **complex ADF pipelines** to integrate batch and real-time data from DB2, Oracle, and flat files into **Azure Synapse and ADLS Gen2**.
- Integrated **CI/CD deployment** of ADF assets using Azure DevOps, ARM templates, and DataOps best practices.
- Designed and supported **real-time and batch data pipelines** using a combination of **SQL, NoSQL**, and distributed processing systems to handle complex and high-volume financial data flows.
- Designed and deployed scalable **GCP Dataflow and BigQuery pipelines** to ingest and process high-velocity transactional data for fraud analytics use cases.

- Built real-time streaming ingestion using **Kafka** integrated with **Cloud Composer (Airflow)** to orchestrate end-to-end ingestion and transformation workflows.
- Designed and implemented complex **ADF pipelines** to ingest DB2 and Oracle data into Azure Synapse for analytics and reporting.
- Developed and scheduled **Azure Functions** for schema validation and triggering metadata updates post-ingestion.
- Implemented automated **unit and integration testing frameworks** using pytest and custom schema validators for data quality enforcement.
- Led **schema translation from DB2 to Snowflake**, including migration of indexes, foreign keys, constraints, and stored procedures (translated into SnowProc-compatible JavaScript logic).
- Designed **Snowflake stored procedures (SnowProc)** to encapsulate business logic for campaign and fraud analytics, enabling parameterized execution and reusability.
- Built **CI/CD deployment pipelines** for ADF assets and parameterized datasets using Azure DevOps and ARM templates.
- Implemented **role-based access control (RBAC)** and **tag-based masking policies** within Snowflake to align with security and compliance guidelines.
- Automated **file validation and metadata tagging** using **Azure Functions** and **Logic Apps integrated with ADLS events**.
- Built an internal **testing and validation tracker** using Python and Snowflake logs to track row counts, column mismatches, and schema conformity across legacy DB2 and Snowflake datasets.
- Designed reusable **DBT tests** for **data quality assertions** such as uniqueness, non-null, and referential integrity across curated layers in **Snowflake**.
- Developed and maintained robust ETL/ELT workflows using **AWS Glue**, **SSIS**, and **StreamSets**, handling both structured and semi-structured formats including **JSON** and **XML**.
- Optimized **Snowflake** queries using clustering, materialized views, and performance tuning for financial analytics.
- Created high-throughput **ETL** pipelines in Talend for transaction ingestion and cleansing from Oracle and SQL Server.
- Integrated **Talend with AWS S3** and Snowflake to automate batch data movement and alerting.
- Enabled version-controlled transformation pipelines using **DBT** integrated with **Git** for team collaboration.
- Integrated Airflow with **GCP Composer** and **AWS Lambda** for hybrid orchestration.
- Developed **BigQuery** queries for cost-optimized risk analytics dashboards and fraud anomaly detection.
- Ingested real-time events via **GCP Pub/Sub** and processed them using **GCP Dataflow for reporting**.
- Built tiered storage in **GCP Cloud Storage** for data archival, backup, and legal retention.
- Designed error-handling logic in Python for real-time ingestion scripts with **Pandas** and **NumPy**.
- Built Spark-based **ETL** in **Scala** for large-scale delinquency data processing in **GCP Dataproc**.
- Developed and optimized multi-threaded financial validators using **Java**.
- Designed real-time **Tableau** dashboards for delinquency rates and portfolio trends.
- Integrated **Snowflake** live connections and parameter filters for interactive **Tableau visualizations**.

Tools Used:

Snowflake, GCP BigQuery, GCP Dataflow, Vertex AI, Talend, AWS S3, DBT, Scala, Java, GCP Pub/Sub, GCP Cloud Storage, Apache Spark, Hadoop, GCP Dataproc, Python, Pandas, NumPy, SQL, SQLAlchemy, DynamoDB, AWS CDK, Apache Airflow, AWS CodeBuild, AWS Athena, Snowflake Streams, Snowflake Tasks, Tableau, Excel, SQL Server, Oracle, AWS Kinesis, AWS Glue, AWS Lambda, AWS Step Functions, AWS Fargate, AWS SageMaker.

PayPal, CA

Jan 2022- Dec 2023

Senior Data Engineer

Functional Role Details:

- Managed **real-time data streaming** to capture, transform, and store **financial transactions for fraud detection and risk management**, ensuring high-throughput and low-latency processing.
- Implemented CI/CD workflows using **GitHub Actions** for testing, deployment, and data drift validation for production pipelines.
- Designed **data quality checks in DBT**, using custom Python-based assertions and alerts, logging results to Slack and JIRA.
- Implemented **BI integration pipelines to Snowflake**, surfacing validated KPIs in **Looker**, powering real-time executive dashboards.
- Migrated pipelines from legacy SSIS to **Azure Data Factory**, implementing dynamic mapping data flows and lookup activities.

- Used **Azure Databricks (PySpark)** to process raw customer and transaction logs, applying business rules and aggregations.
- Orchestrated daily and hourly ADF pipelines using **trigger-based logic**, integrating with Synapse and Snowflake targets.
- Designed **Power BI datasets and semantic models** connected to Azure SQL for real-time decision-making dashboards.
- Used **Snowflake tasks, streams, and SnowProc** to implement incremental data refresh and late-arriving data handling logic.
- Led **SQL Server to Azure migration**, ensuring **schema optimization, indexing, and data consistency**, and enabling **seamless integration with cloud-based analytics platforms**.
- Designed an **archival solution** using Snowflake external stages and lifecycle policies to retain and separate cold data from frequently accessed analytics tables.
- Integrated testing automation for historical loads using Python scripts to validate data integrity over multiple load cycles and prevent duplication.
- Built API integrations using Python to validate third-party fraud inputs and enrich customer profiles in Snowflake.
- Optimized **database performance for high-volume payment transactions**, enhancing **query efficiency, indexing**
- Utilized **AWS Glue, S3, and Redshift** to develop and maintain large-scale ETL/ELT pipelines handling structured and semi-structured data.
- Ensured data consistency and reliability using **JSON/XML validation**, schema enforcement, and **automated reconciliation jobs** via Airflow.
- Built secure and scalable infrastructure using **Terraform, Docker, and CI/CD pipelines**, while ensuring **DevOps best practices** for version control and deployment via **Git and Jenkins**.
- Delivered analytics dashboards and reporting datasets through **Looker and Tableau**, supporting executive decision-making and compliance tracking.
- Migrated **ETL logic** from on-prem SQL Server to Snowflake, enhancing scalability and cost-efficiency.
- Stored pre-computed embeddings in **BigQuery** to experiment with hybrid search for support ticket summarization.
- Applied security and access controls in Snowflake, aligning with PayPal's data governance policies.
- Used **Terraform** to provision **Redshift, RDS, and Glue resources** in development and production environments.
- Developed **real-time streaming pipelines** to handle **market data and payment transactions**, ensuring **low-latency processing**.
- Optimized **Apache Spark and Hadoop clusters**, improving **batch processing efficiency for financial risk models** and credit scoring.
- Led **SQL Server to Azure migration**, ensuring **schema optimization, indexing, and data consistency**, and enabling **seamless integration with cloud-based analytics platforms**.

Tools Used:

AWS (Firehose, SQL Server, RDS, CDK, Lambda), Vertex AI, Azure (Synapse, ADLS), Snowflake, DBT, PostgreSQL, SQL Server, Tableau, Flask, Pandas, NumPy, Pyarrow, Bitbucket, Flask, Pandas, NumPy, Snowflake, Azure Synapse, ADLS (Azure Data Lake Storage), PostgreSQL, DBT, Python, AWS CDK, AWS Quicksight, AWS RDS, Docker, AWS Glue, Terraform, IAM, VPC, Subnets, Route Tables, PCI DSS, Tableau, Bitbucket, Pyarrow, Scikit-Learn.

Wells Fargo, IND

Sep 2018- Dec 2021

Senior Data Engineer

Functional Role Details:

- Designed and implemented real-time streaming pipelines to capture, transform, and store financial transactions, ensuring efficient fraud detection and risk assessment in high-volume payment processing.
- Migrated critical data pipelines from SQL Server to **Snowflake** using **ADF and Databricks**, ensuring secure, validated migrations.
- Created reusable templates and macros for BigQuery ML training/evaluation, standardizing ML practices across regulatory and reporting departments.
- Created **data quality rules using Python and DBT**, with daily quality KPIs published to **Power BI** dashboards.
- Partnered with enterprise governance to catalog assets in **Informatica Axon**, enabling stewardship workflows, ownership assignment, and traceable lineage.
- Created enterprise-scale **ADF pipelines** for ingesting, cleaning, and enriching financial data from SQL Server and Oracle sources.

- Built **reusable Databricks notebooks** in Python/Scala for real-time streaming data preparation and feature extraction.
- Designed partitioned and secure **ADLS data lakes** for customer, account, and transaction-level data, following Lakehouse architecture principles.
- Implemented **Azure Functions** to trigger Power BI dataset refreshes and run control table audits.
- Supported **data quality initiatives** by integrating ADF with custom Python validation logic and alerting frameworks.
- Built robust **Spark (Scala and PySpark)** applications on **Hadoop** and later migrated workloads to **Snowflake** and **Azure SQL Data Warehouse** for improved scalability and performance.
- Developed custom **ETL/ELT frameworks** using **SSIS, StreamSets**, and **Azure Data Factory (ADF)** to move, cleanse, and structure regulatory data.
- Implemented **CI/CD workflows** using **Git, Jenkins**, and **Terraform**, supporting release automation, environment provisioning, and infrastructure-as-code practices.
- Produced executive-facing reports and analytics dashboards via **Tableau** and **Looker**, consolidating data quality metrics, audit outcomes, and fraud alerts.
- Led **SQL Server to Snowflake migration**, ensuring **seamless data transformation and lineage tracking**.
- Designed **Snowflake-based ELT workflows**, improving **data consistency and reducing redundant processing**.
- Refactored data transformation layers using **DBT** for better modularity and auditability.
- Automated schema testing and documentation with **DBT snapshots and sources**.
- Integrated **DBT jobs** with version control and CI/CD workflows.
- Used **Terraform** to provision **Redshift, RDS, and Glue** resources in development and production environments.
- Automated **IAM policies and S3 bucket** configurations using reusable Terraform modules.
- Deployed secure and versioned infrastructure using remote state and **CI/CD integration in Terraform**.
- Designed high-performance **Snowflake** data models, improving query efficiency and reducing storage costs.
- Implemented **Snowflake Streams and Tasks**, enabling real-time data ingestion and transformation for fraud detection.

Tools Used: AWS Firehose, SQL Server, Flask, Pandas, NumPy, Snowflake, Azure Synapse, ADLS (Azure Data Lake Storage), PostgreSQL, DBT, Python, AWS CDK, AWS Quicksight, AWS RDS, Docker, AWS Glue, Terraform, IAM, VPC, Subnets, Route Tables, PCI DSS, Tableau, Bitbucket, Pyarrow, Scikit-Learn.

HSBC, IND

July 2015- Sep 2018

Data Engineer

Functional Role Details:

- Developed scalable data processing pipelines leveraging **PySpark and Scala**, utilizing in-memory computing and RDD transformations to efficiently process financial transaction data.
- Developed and optimized Scala and Java-based data processing applications, ensuring efficient handling of structured and unstructured financial datasets.
- Reverse-engineered complex DB2 schemas used in AML systems and rebuilt them in Snowflake with appropriate **data types, constraints, and stored procedures (SnowProc)**.
- Created **automated trackers for migration validation**, highlighting column-level anomalies, transformation mismatches, and null/zero ratio shifts during migration from DB2 to Snowflake.
- Built testing pipelines with **differential queries** to assess drift between DB2 and Snowflake tables after historical batch loads.
- Authored **unit test cases for data flow validation**, collaborating with audit teams and SMEs to simulate regulatory query scenarios and validate final data marts.
- Employed **Snowflake masking policies** for PII and applied granular access controls for various audit roles.
- Managed **data governance catalogs and lineage** for regulatory datasets using **Informatica EDC** and Hive metadata.
- Tracked weekly Jira tickets and created audit trails for each table migration or schema change in Hive.
- Optimized **regulatory reporting frameworks** using **GCP BigQuery**, reducing data retrieval times by 35%.
- Automated **data ingestion workflows** in **GCP Dataflow**, ensuring **seamless integration of financial data sources**.
- Implemented **event-driven architectures** using **GCP Pub/Sub**, improving **real-time banking alerts and notifications**.
- Migrated **on-premises financial data** to **GCP Cloud Storage**, enhancing **scalability and cost efficiency**.
- Utilized **GCP Dataproc** to **process large-scale financial records**, reducing **batch processing time for loan approvals**.
- Configured and fine-tuned Apache Spark clusters on **Hadoop YARN**, optimizing executors, memory distribution, and parallelism settings to enhance large-scale banking data analytics.

- Built **Tableau visualizations** to provide insights into **data integrity, consistency, and business intelligence trends**.
- Implemented data partitioning and bucketing strategies in **Apache Spark SQL and Apache Hive**, minimizing shuffle operations and improving query performance for regulatory compliance reports.
- Automated and orchestrated big data workflows using **Apache Airflow**, ensuring efficient scheduling, dependency management, and monitoring of complex **ETL** pipelines handling customer financial records.
- Integrated **Apache Kafka** for real-time data streaming, enabling high-throughput message processing for fraud detection and real-time transaction monitoring.
- Leveraged the **Hadoop** ecosystem including **HDFS, YARN, Apache Pig, and Apache Hive**, optimizing data storage, retrieval, and distributed processing for large-scale banking datasets.
- Developed **ETL** pipelines using **Apache Spark and Hadoop MapReduce**, automating financial data ingestion, transformation, and processing to support risk analysis and credit scoring.
- Enhanced query performance in **Apache Hive and Presto** by implementing advanced data partitioning techniques, significantly reducing data scan times in regulatory audits.

Tools Used: GCP BigQuery, GCP Dataflow, GCP Pub/Sub, GCP Cloud Storage, GCP Datapro, Apache Spark, Apache Hadoop, HDFS, YARN, Apache Pig, Apache Hive, Apache Impala, Apache Presto, Apache Flink, Apache Storm, Apache Kafka, Java, Apache Airflow, Scala, PySpark, Hadoop MapReduce.

United Health Group, IND

Aug 2013- June 2015

Data Engineer

Functional Role Details:

- Designed and implemented **Hadoop-based data pipelines** to ingest and process high-volume healthcare claims, eligibility, and enrollment data from multiple source systems.
- Used **HDFS** for distributed storage of structured and semi-structured healthcare datasets, applying optimal block size and replication strategies.
- Developed **Hive scripts and tables** for transforming and querying large datasets related to patient encounters, clinical codes, and pharmacy data.
- Utilized **Pig Latin scripts** for cleansing and enriching raw data sets, enabling downstream use in member utilization and cost analysis reports.
- Integrated **Sqoop** to efficiently import bulk data from enterprise RDBMS systems (Oracle, Teradata) into HDFS and Hive for analytical consumption.
- Scheduled and orchestrated batch workflows using **Apache Oozie**, automating the execution of Hive, Pig, and shell scripts for nightly processing.
- Developed custom **MapReduce jobs** in Java for specific aggregation logic involving patient risk scoring and preventive care metrics.
- Leveraged **Flume** to stream log files from clinical data services into Hadoop clusters for centralized diagnostics and monitoring.
- Queried healthcare performance indicators using **Impala** for fast, interactive analytics dashboards consumed by the clinical analytics team.
- Tuned cluster performance using **YARN resource manager**, ensuring optimized job execution, node stability, and memory utilization during peak load.
- Worked with **Apache HBase** to store and retrieve time-sensitive, low-latency claims data and eligibility lookup records for real-time access in analytics dashboards.
- Used **Apache Avro** for serializing structured healthcare records and schema evolution, supporting seamless data exchange between batch and real-time jobs.
- Integrated **Apache ZooKeeper** for distributed coordination and synchronization between services within the Hadoop cluster, improving stability and high availability.
- Explored **Apache Mahout** for initial experimentation with collaborative filtering models, focusing on patient similarity and early readmission risk scoring use cases.
- Acted as the primary liaison between **engineering and healthcare analytics teams**, translating complex data processing logic into clear business terms and delivering technical walkthroughs to non-technical stakeholders.
- Resolved critical data latency issues by redesigning ingestion workflows in Hadoop, improving ETL throughput by 40% and ensuring timely access to claims and eligibility data for downstream analytics.

Tools Used: Hadoop, HDFS, Hive, Pig, Sqoop, MapReduce, Oozie, Flume, Impala, YARN, Ambari, HBase, Avro, Zookeeper, Mahout